

A Trace-Based Validation Protocol for Seed-Matched Evaluations of Black-Box Game-Playing Agents

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Recording the same seed for two game-playing agents establishes a matched schedule, but it does not show that either execution repeats or that the agents receive equal random quantities for the same semantic events after their paths diverge. We present an executable, fail-closed protocol that binds artifact identity to the value passed at the observable engine boundary, checks row-level schedule parity, compares identical-arm trace projections across repeat and worker contexts, audits stochastic sources, and applies a frozen rule that admits, downgrades, or suppresses statistical claims. A self-contained synthetic suite exercises five controlled modes and shows that an event-keyed construction restores alignment for all five logged shared events after a stateful draw shift aligned only one. In the restricted-engine case study, the historical repeated-control audit disagreed on 210/2,800 available win/draw records and 458/2,800 records after decision count was added, so the planned historical comparison was suppressed. A prospectively frozen deterministic preflight then found zero disagreements across 1,000 arm-seed-condition units and 3,000 executions. A separate timed-search stress test found 99/200 trace-projection disagreements (49.5%; empirical 95% cluster-resampling interval [42.5%, 56.5%]). The admitted factorial covered 2,000 fixed-schedule units; its total contrast was +0.55 percentage points (empirical 95% paired-resampling interval [-2.05, +3.15]), which is compatible with effects in either direction and does not establish equivalence. The principal limitation is that the restricted engine exposes neither semantic event identifiers nor event-keyed streams, so its case-study results cannot establish cross-arm event alignment.

I. INTRODUCTION

Reusing random inputs can sharpen a comparison when two stochastic systems respond similarly to the same sources of variation. Common-random-number (CRN) theory makes that gain conditional on the construction and on properties such as system structure and event timing, rather than on the reuse of an integer alone [1]. Streams and substreams are established ways to organize synchronized simulation and independent replications [2]. A recent paired-seed preprint likewise studies the conditional precision gain when seed-level outcomes are favorably correlated [3]. These results motivate careful pairing; they do not turn a seed column into evidence of a particular coupling.

The gap matters for black-box game-agent evaluations. A researcher may be able to hash a restricted simulator, call a seeded entry point, freeze an opponent schedule, and record public actions, yet remain unable to inspect the simulator’s generator or assign stable identifiers to random events. Separate executions can also encounter clock-bounded search, process-local state, or different queue histories. Under those conditions, the same recorded seed can name a matched design without establishing repeatable execution or semantic alignment across arms.

This article addresses one question: when two agents are evaluated under the same recorded seed, what evidence is required before the result can legitimately be

analyzed as a paired comparison? We integrate four distinct evidence layers—schedule matching, within-arm execution repeatability, cross-arm event alignment, and statistical claim admission—into an executable validation protocol for restricted black-box settings. The protocol records artifact and seed-boundary identity, audits every scheduled row, compares a declared trace projection in A/A and repeat/worker tests, inventories stochastic sources, and enforces a prospectively frozen claim map. It fails closed: a later point estimate or test cannot repair a missing gate.

The contribution is the operational integration of a declared recorded-trace projection, a self-contained conformance suite, and an executable admission map. We do not claim that same seeds can fail, CRN requires synchronization, A/A controls are new, fixed-work search is new, event-keyed randomness is new, or reproducibility checklists are new. The restricted game environment is a case study, not the title-level contribution, and no competitive-strength claim is made.

II. RELATED WORK AND CONTRIBUTION BOUNDARY

CRN conditions, stream design, paired-seed evaluation, and event-keyed randomness already cover the central stochastic-computation ingredients [1–4]. The Sharma and Buffalo papers are arXiv preprints, not verified peer-reviewed publications. The latter directly motivates stable event keys as a white-box repair for stateful draw shifts, but it does not validate the event ontology or the black-box engine used here.

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The broader empirical literature distinguishes reproducible code from reproducible findings, emphasizes explicit units and sources of variation, and recommends uncertainty-aware reporting across heterogeneous tasks [5–8]. Simulation verification and validation distinguish implementation checks from operational validity [9]. Metamorphic testing has been applied both to scientific software without a convenient single-run oracle and to simulation verification and validation [10, 11]. A/A experiments are also an established diagnostic in randomized online experimentation [12]; the identical-artifact trace repetition used here adapts that diagnostic idea, rather than importing its sampled-user inference. Wall-clock allocation is already an algorithmic design concern for Monte Carlo tree search [13]. That literature supports auditing time-bounded search but does not identify the cause of a particular observed divergence.

The boundary in Table I is therefore deliberate. The protocol does not replace simulator validation, prove favorable seed-level covariance, or guarantee external generalization. It supplies a black-box workflow that turns prior component methods into an explicit decision about which paired claim, if any, the recorded evidence permits.

III. DEFINITIONS

A. Schedule matching

A schedule is the prospectively declared set of evaluation rows. Two arm rows are schedule matched when all pairing variables required by the protocol agree: the value passed at the observable engine boundary, opponent, actual order, physical seat, environment, decision limit, and other fixed configuration. This is a property of design records. It says that a matched comparison was attempted; it does not say that either execution is repeatable.

B. Execution repeatability

Execution repeatability is a within-arm property. For the same artifact, seed condition, and declared execution context, the compared execution record must agree across fresh runs. Here the strongest trace-bearing record is a canonical sequence of public-observation hashes, acting side, selected action, and terminal record. The implementation compares the SHA-256 digest and byte count of that complete recorded projection. It does not retain raw public observations, opaque search snapshots, hidden simulator state, or semantic random-event identifiers. Thus “complete recorded trace projection” always means complete only within this declared projection.

Trace agreement is stronger than outcome agreement because different action sequences can end with the same winner. Conversely, a digest is not a proof that omitted state agreed. A single required mismatch rejects exact

repeatability on the exercised schedule; a pass remains scoped to the tested artifacts, projection, and contexts.

For a stress test with N seed-condition clusters, let H_{ir} be the unitless trace digest for cluster i under execution profile r . The trace-disagreement proportion is

$$\hat{q} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(|\{H_{ir} : r \in \mathcal{R}\}| > 1). \quad (1)$$

Here $\mathcal{R}$ is the prescribed set of profiles, the indicator is one when at least two profile digests differ, and $\hat{q}$ is a unitless proportion of seed conditions. In plain language, the equation counts seed conditions whose repeated traces disagree. If two of five clusters disagree, $\hat{q} = 2/5 = 0.40$. The independent executable example is recorded in `tests/test_equations.py`.

C. Semantic event alignment

Semantic event alignment is a cross-arm property. After two policies branch, every shared modeled exogenous event in the declared ontology must retain a stable identifier and receive the same random quantity. Logged event/value pairs, event-keyed counter-based randomness, or a validated assignment of event-specific streams can provide this evidence [2, 4]. Exact within-arm repetition cannot establish it: identical policies ordinarily follow the same path and consume the same stateful sequence, while different policies can shift later draws.

An event-alignment pass is conditional on the ontology, marginal random distributions, dependence structure, and logged event classes. The restricted engine lacks these event identifiers, so its evaluations cannot establish this property.

D. Statistical admission

Statistical admission is a decision fixed before outcome inspection. It maps the evidence achieved to an estimand, analysis unit, uncertainty procedure, target population, and exact wording. Schedule evidence can authorize only a matched-schedule description. Repeatability plus a bounded source audit can, under the frozen rule, authorize an implementation-conditional seed-matched estimate. Event-aligned wording additionally requires semantic event evidence. A failed required gate suppresses or downgrades the affected comparison even if its interval excludes zero.

Figure 1 summarizes the non-implications among these definitions.

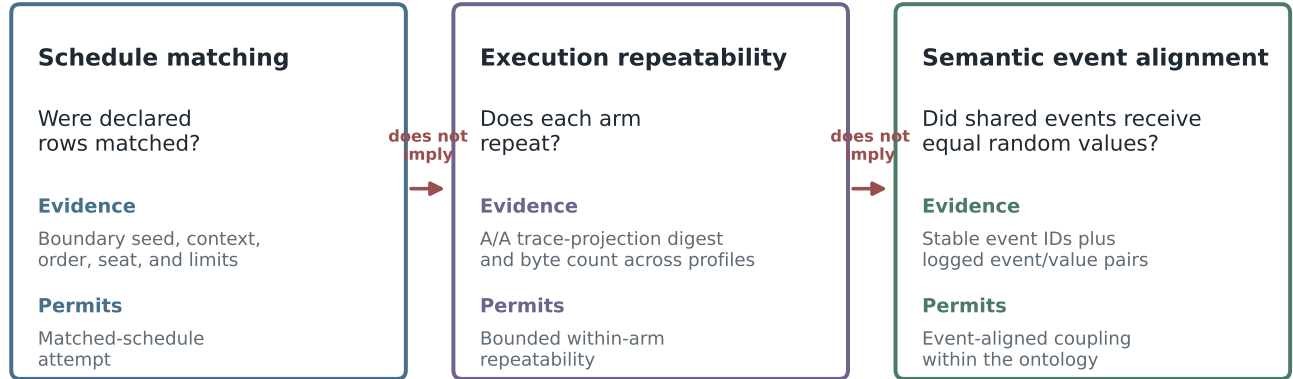
IV. TRACE-BASED VALIDATION PROTOCOL

The protocol solves an admission problem, not a policy-selection problem. Its analysis unit is defined at each

TABLE I. Relationship to prior methods. The present contribution is their operational integration for black-box agent evaluation, not priority over the component ideas.

Prior area	Established contribution	Role here
Common random numbers	Conditions for covariance and variance reduction; synchronization depends on model structure and event timing [1]	Motivates separating a matched integer from an evidenced coupling.
Streams and substreams	Organized independent replications and synchronized streams [2]	White-box design option; stream labels alone do not prove semantic assignment.
Paired-seed evaluation	Precision can improve when seed-level outcomes are favorably correlated [3]	Supports pairing when its implemented relationship is validated.
Event-keyed randomness	Stable event keys prevent draw shifts within a declared event ontology [4]	Implemented in the synthetic repair; unavailable inside the restricted engine.
Reproducibility and RL evaluation	Variation, units, estimation, and transparent reporting are essential [5–8]	Supplies the broader empirical-design boundary.
Software and simulation testing	A/A, metamorphic relations, and verification/validation test implementations and models [9–12]	Informs trace invariants and fail-closed conformance checks.
Work versus time budgets	Search time allocation is established prior art [13]	Justifies auditing clock-bounded search without claiming its invention.

Three distinct validation questions



Evidence accumulates left to right; later claims require additional observations rather than stronger wording.

FIG. 1. Schedule matching, within-arm execution repeatability, and cross-arm semantic event alignment answer different questions and require different evidence. Each claim is limited to the declared schedule, trace projection, execution contexts, and event ontology. Source data are in `source_data/figure_1_distinctions.json`.

stage: artifacts at identity, arm-paired rows at schedule parity, arm–seed conditions at preflight, seed-condition clusters containing all execution profiles at stress testing, and paired game units within frozen opponent-by-order strata at factorial estimation. Opponent and worker profiles are fixed execution contexts, not random samples from a larger population.

A. Stages 1–3: identity, boundary seed, and schedule

Stage 1 hashes the simulator, policies, opponents, protocol, evaluator, and configuration. The problem it solves

is artifact ambiguity: a result without identified bytes cannot be attributed to the declared system. A hash match does not validate behavior. Missing or drifting artifacts invalidate the affected run.

Stage 2 records the scheduled host-language seed and the exact integer passed at the observable engine boundary. This is needed because an adapter may narrow, reduce, or otherwise convert a value. The full schedule is converted before execution and checked for unintended collisions and overlaps. The engine’s internal handling remains unobserved; the protocol never calls the boundary value an internally consumed seed.

Stage 3 checks equality and completeness for every required schedule field. The output is a row-level parity

TABLE II. Protocol stages, required evidence, strongest permitted statement, and failure action. Every pass is local to the tested artifacts, schedule, trace schema, and execution contexts.

Stage	Evidence	Permitted statement	Failure action
1 Artifact identity	Hashes for engine, policies, opponents, protocol, code, and configuration	The evaluated bytes are identified.	Quarantine missing or drifting artifacts.
2 Seed namespace	Scheduled seed, exact boundary value, conversion rule, and collision audit	The claimed boundary seed was supplied and units remain distinct.	Correct the adapter or schedule and rerun.
3 Schedule parity	Row equality for opponent, order, seat, seed, environment, and limits	A complete matched schedule was attempted.	Reject unequal or incomplete pairs.
4 Identical-arm parity	Separate A/A executions compared through the digest and byte count of the recorded trace projection	The compared fields repeat on exercised A/A trajectories.	Suppress the affected paired comparison.
5 Repeat/worker parity	Fresh repeats, worker counts, enqueue orders, and process lifecycles	Within-arm traces repeat in tested contexts.	Freeze a passing context or model run variation.
6 Source audit	Bounded Python source-pattern scan; binary internals remain unassessed	Candidate mechanisms and the uninspected boundary are recorded.	Control, dynamically test, or narrow the estimand and wording.
7 Event alignment	Stable event identifiers and equal values for shared exogenous events	Event-aligned coupling for the logged ontology.	Retain at most bounded seed-matched wording.
8 Statistical admission	Frozen map from gates to estimand, unit, uncertainty, population, and wording	Only the prespecified supported claim class.	Admit, downgrade, or suppress automatically.

report and schedule fingerprint. A missing cell, unequal field, error, or silent deletion invalidates the comparison. A historical data set that lacks fields required by the current protocol can support agreement only for its recorded projection, not a retrospective pass of the full stage.

B. Stages 4–6: trace repetition and source audit

Stage 4 launches byte-identical arms separately and compares their records in increasing strength: outcome, errors, decision count, and the digest plus byte count of the recorded trace projection. This A/A adaptation detects implementation or execution differences that an outcome check can miss. Its assumptions are that the serialization is canonical, the digest comparison is implemented correctly, and all fields relevant to the claim are declared. It does not establish cross-arm coupling.

Stage 5 prespecifies fresh serial repeats, worker counts, enqueue orders, and pool lifecycles. Repeats within one seed condition share the same planned inputs and are not independent observations. Worker profiles test execution contexts; they are not sampled hardware replicates. The output is a cluster-level trace-parity report. Any required mismatch rejects exact parity for at least that exercised seed condition.

The implemented Stage 6 is a bounded Python AST and source-pattern scan of package trees. It inventories explicit random interfaces, clock or deadline patterns, concurrency APIs, module-level state, and native pointer or handle patterns. A static hit identifies a mechanism for dynamic testing but does not show that the branch

executed or caused a mismatch; a zero-hit scan does not prove determinism. The scanner does not inspect binary-engine internals or classify hardware and external state. The stage is invalid if the inventory or its scope is missing, and it cannot identify a unique cause.

C. Stages 7–8: alignment and claim map

Stage 7 compares stable identifiers and random values for shared events after policy branching. The output names the covered event classes and any misalignment. Because the case-study engine exposes neither event keys nor independent event streams, the stage is unavailable there. This blocks event-aligned, counterfactual, and full-CRN wording but does not erase the lower-level evidence.

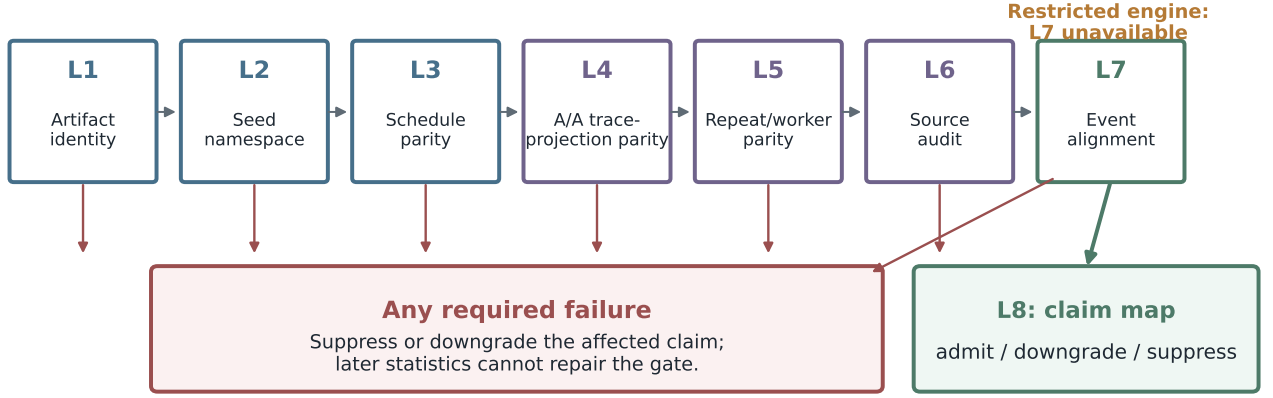
Stage 8 executes the frozen admission map shown in Fig. 2. The map may admit a bounded seed-matched estimate, downgrade its interpretation, or suppress it. It cannot use a favorable result to waive an earlier requirement. The analyzer also rejects nonfinite results, schema or commit changes, sample-size drift, missing strata, and requests for suppressed output.

D. Paired estimand and resampling

For binary outcome Y (win = 1, nonwin = 0), the paired difference for unit i is

$$d_i = Y_i^{(I)} - Y_i^{(C)}. \quad (2)$$

Trace-based protocol and claim admission



Passes are scoped to the tested artifacts, trace schema, seed schedule, and execution contexts.

FIG. 2. Validation and admission flow. Levels 1–7 accumulate scoped evidence. Any required failure invokes the prospectively frozen failure action; Level 8 cannot repair it. Source data are in `source_data/figure_2_admission_flow.json`.

The superscripts denote intervention and control; each Y and d_i is unitless, with $d_i \in \{-1, 0, 1\}$. The mean is reported as a percentage-point win-rate difference. In ordinary language, d_i records which arm won for a matched unit. If intervention wins and control does not, $d_i = 1 - 0 = 1$. This identity is independently tested in `tests/test_equations.py`. It does not by itself show that the paired outcomes are event aligned.

For K frozen strata with n_k paired units, bootstrap replicate b is

$$\bar{d}^{*(b)} = \frac{1}{K} \sum_{k=1}^K \frac{1}{n_k} \sum_{j=1}^{n_k} d_{k, I_{bkj}}, \quad (3)$$

$$I_{bkj} \sim \text{Uniform}\{1, \dots, n_k\}.$$

The index I_{bkj} resamples whole paired-unit identifiers within stratum k ; arms are never resampled separately. The result is a unitless mean, reported in percentage points. With one stratum, differences $[1, 0, -1]$, and sampled indices $[1, 1, 2]$, the replicate is $(1 + 1 + 0)/3 = 2/3$. The equation assumes the frozen within-stratum empirical distribution is the object being resampled. Its percentile interval is an algorithmic stability summary for the realized battery, not a population confidence interval over new opponents, training runs, or execution contexts [14].

E. AI-assisted research methods

OpenAI Codex, using a GPT-5-family model for which the exact deployed snapshot was not exposed, pro-

vided substantive research assistance under human direction. Recorded tasks included protocol reasoning, literature synthesis, code generation and debugging, simulation orchestration, statistical-analysis code, deterministic visualization, and adversarial auditing. The human author set the scientific question, froze the acquisition protocol, retained authority over experiments and claims, and must approve the final paper. Verification included artifact hashes, strict schemas, row-level reaggregation, independent equation tests, targeted software tests, comparison against primary literature, a one-command reproduction run, compilation, and page-image inspection. The machine-readable activity record is `supplement/AI_USE_LOG.csv`. No generative-image system was used; all figures are deterministic plots produced by the checked-in artifact builder.

V. SELF-CONTAINED SYNTHETIC CONFORMANCE SUITE

The standalone standard-library suite supplies an inspectable review-package test oracle without the restricted game engine. Five fixed modes exercise distinct non-implications: a clean deterministic mode, a stateful draw shift, an injected clock-budget mode, an injected process-state mode, and a seed conversion mode. Injected profiles make the fixtures reproducible; they are demonstrations of mechanisms, not claims that every external simulator contains them.

The seed-conversion fixture repeats within arm while failing namespace integrity. The clock and process fixtures fail trace repetition and execution- context checks.

The stateful draw-shift fixture repeats exactly within each arm but aligns only one of five shared events after one arm consumes an extra draw. Deriving random quantities from the base seed and semantic event key aligns all five logged shared events. This repair is established only for the fixture’s declared event ontology and source distribution.

The suite writes complete JSON evidence, a level-by-mode CSV, a JSON Schema, and a SHA-256 manifest. Within the companion review package, generation and verification are exposed as `python -m pevl_bench generate` and `python -m pevl_bench verify`. Checked tests validate the five fixed mode decisions, event-keyed repair, schema rules, and manifest-tamper rejection.

VI. RESTRICTED GAME-AGENT CASE STUDY

The case study uses a seeded tournament-engine binary, four fixed policy packages, and third-party opponent packages that cannot be redistributed under the currently established rights. Policies are described only as control, representation intervention, training intervention, and combined intervention. Opponents are grouped as deterministic or timed-search execution contexts. The scientific target is the validation protocol; policy quality and game strategy are outside scope.

The bounded source audit verified frozen digests for all 13 inventoried artifacts and applied the Python source-pattern scan to 11 package trees. The 2 engine binaries were hash checked but not source assessed. These records identify candidate mechanisms and an explicit uninspected boundary; they do not prove determinism or a causal mechanism.

The historical study scheduled seven opponents, two actual orders, and 200 seed conditions per opponent-order cell. Candidate and control rows recorded the same requested seed, opponent, order, and seat, but ran as separate process-pool tasks. The historical files do not contain the complete field set required for a current Stage-3 pass and did not capture the later trace projection. The strongest retrospective check is therefore equality of an available-record projection across three separately executed control arms.

That projection disagreed on 210/2,800 units when it included win and draw fields and on 458/2,800 units after decision count was added. Both mismatch sets were confined to two timed-search opponent packages. Under the frozen rule, the historical factorial contrasts were not estimable and were suppressed. The mismatches are evidence of failed repeatability for those records, not proof that wall-clock timing alone caused every divergence. Removing the two opponents after inspecting the failure would have changed the target population and was not used as a rescue analysis.

VII. PROSPECTIVE VALIDATION RESULTS

The protocol content first appears at commit 803257f1, and the retained result artifacts appear in descendant commits. Repository history establishes that artifact ordering; it cannot independently prove when a human inspected uncommitted files. Artifact hashes, the boundary seed conversion, schedule rows, trace schema, execution profiles, sample sizes, bootstrap seeds, and admission actions were fixed at protocol commit 803257f1. The prospective boundary-seed audit found all scheduled values in range and distinct, with no overlap with the converted historical namespace.

A. Deterministic preflight

The preflight covered four arms and five prospectively frozen deterministic opponents. Each arm-opponent job contained both actual orders, 25 seeds per order, two fresh serial profiles, and one eight-worker profile. Across 1,000 arm-seed-condition units and 3,000 execution trajectories, the digest and byte count of the complete recorded public-observation-hash/action stream, terminal outcome, errors, and decision count had 0 mismatches. This passed the frozen repeatability gate and admitted factorial acquisition. It did not establish behavior on untested hardware or loads, and it did not establish cross-arm event alignment.

B. Timed-search stress test

The separate diagnostic retained the two timed-search opponents rather than silently excluding them. Each seed-condition cluster contained four profiles: serial forward, serial reverse, parallel forward, and parallel reverse. Of 200 clusters and 800 executions, 99 clusters had a trace-projection disagreement (49.5%; empirical 95% whole-cluster-resampling interval [42.5%, 56.5%]). Outcomes disagreed in 47 clusters, decision counts in 93, and error records in 0. All 99 localized earliest differences occurred on recorded opponent-action events. This rejects exact repeatability for at least one exercised seed condition and identifies plausible clock or process mechanisms; it does not prove a unique cause.

VIII. GATED FACTORIAL DEMONSTRATION

The factorial crossed a representation flag and a separately trained policy update, producing four fixed packages: C1 (neither), C2 (representation only), C3 (training only), and C4 (both). Each cell used the same five-opponent, two-order schedule with 200 paired units per stratum. Factorial acquisitions recorded schedule, boundary-seed, outcome, error, and decision-count fields; they did *not* record trace digests. Their admission

Synthetic conformance suite: detected failure and admission

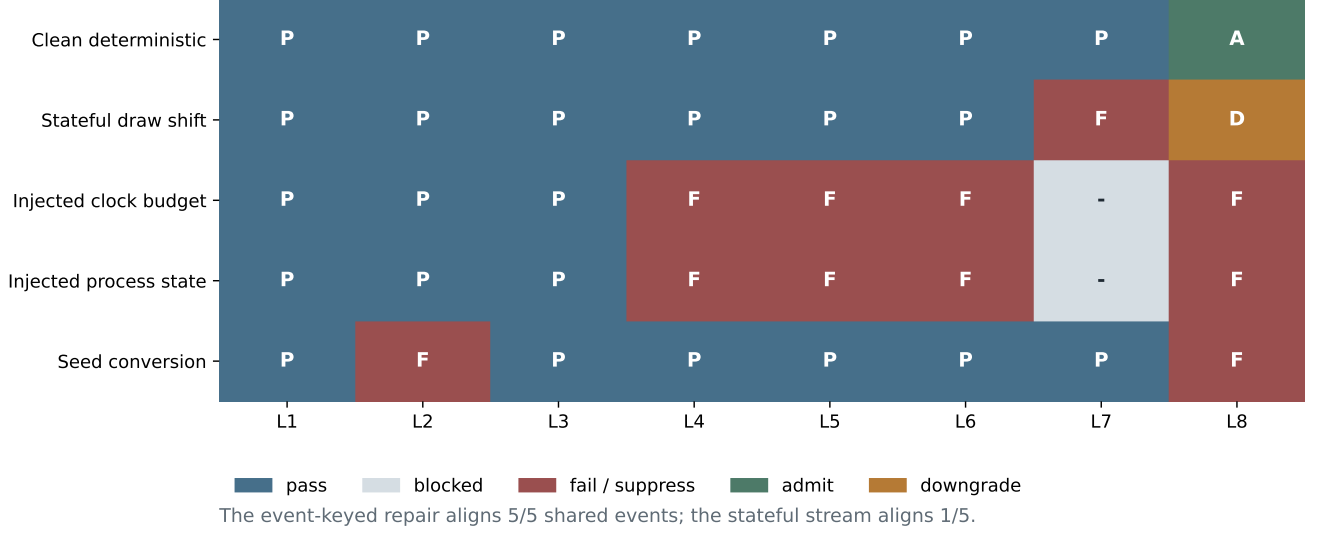


FIG. 3. Synthetic failure-detection and admission matrix. P denotes pass, F fail, A admit, D downgrade, and a dash a blocked gate. The suite demonstrates known logical relations; it does not estimate failure prevalence in external systems. Source data are in `source_data/figure_3_synthetic_matrix.csv`.

TABLE III. Prospective validation results. Trace-digest fields were captured for preflight and stress executions but not for factorial outcome rows.

Stage	Units/clusters	Executions	Trace-digest disagreement	Outcome disagreement	Admission
Deterministic preflight	1,000	3,000	0	0	Admit factorial acquisition for the frozen five-opponent scope.
Timed-search stress	200	800	99	47	Reject exact repeatability for at least one exercised seed condition.
Factorial repeated control	2,000 per cell	12,000 games	—	0	Admit bounded seed-matched contrasts from the captured record; Level 7 remains unavailable.

gate was therefore the prespecified repeated-C1 available-record audit, not a claim of factorial trace parity.

Let $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}$ be the unitless win rates for C1, C2, C3, and C4, respectively, where the first subscript marks representation and the second training. The four reported percentage-point contrasts are

$$\begin{aligned}
 T &= \mu_{11} - \mu_{00}, \\
 R &= \frac{1}{2}[(\mu_{10} - \mu_{00}) + (\mu_{11} - \mu_{01})], \\
 G &= \frac{1}{2}[(\mu_{01} - \mu_{00}) + (\mu_{11} - \mu_{10})], \\
 J &= \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}.
 \end{aligned} \tag{4}$$

T is the total fixed-package intervention contrast, R the average representation contrast, G the average training contrast, and J the interaction. All are unitless before multiplication by 100. With cell rates (0.50, 0.52, 0.51, 0.54), the four contrasts are (0.04, 0.025, 0.015, 0.01). The signs and cell directions are tested in `tests/test_equations.py`. These algebraic contrasts do not isolate a mechanism unless the design and execution assumptions hold.

The repeated-control audit had 0 mismatches, so the frozen rule admitted the finite-schedule seed-matched analysis of 2,000 common units per cell (12,000 engine games across the four cells and separately executed controls). The total contrast was +0.55 percentage points with paired-resampling interval $[-2.05, +3.15]$. The representation, training, and interaction contrasts were +0.95, -0.40, and +1.30 percentage points, respectively; all four intervals span zero. These estimates are compatible with effects in either direction on the frozen schedule. They do not prove equality, equivalence, policy superiority, or event-aligned counterfactual effects.

IX. INTERPRETATION AND PERMITTED CLAIMS

The evidence supports four bounded conclusions. First, matching recorded seeds does not by itself establish within-arm repeatability or cross-arm event alignment. Second, the historical available-record mismatch inval-

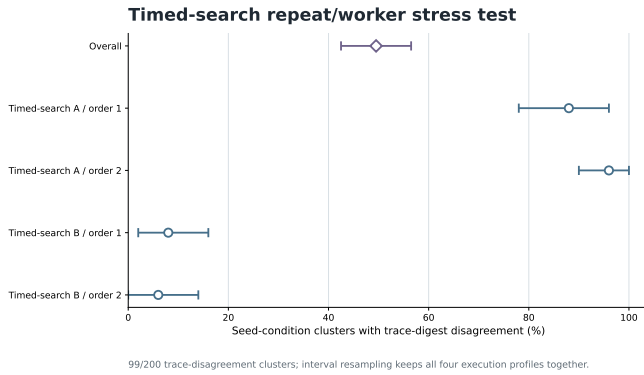


FIG. 4. Trace-projection disagreement in the timed-search stress test. Intervals resample whole seed-condition clusters, preserving all four execution profiles. Any mismatch rejects exact parity on the exercised schedule. Neutral labels replace package names. Source data are in `source_data/figure_4_timed_search.csv`.

TABLE IV. Admitted factorial contrasts. Values are percentage-point win-rate differences with 95% paired-resampling intervals.

Contrast	Estimate	Low	High
Total intervention	+0.55	-2.05	+3.15
Representation	+0.95	-1.08	+3.00
Training	-0.40	-2.03	+1.25
Interaction	+1.30	-1.85	+4.45

icated the planned historical paired decomposition under its frozen rule. Third, the prospective deterministic preflight established repeatability for its exact artifacts, trace projection, schedule, and execution contexts, while the timed-search stress test rejected exact parity on exercised seeds. Fourth, the admitted factorial produced null-compatible finite-schedule estimates without establishing equivalence or superiority.

The synthetic suite supports a stronger event-alignment statement only inside its logged ontology. The restricted engine cannot establish that level. Its factorial is therefore described as seed matched and implementation conditional, not as a fully coupled counterfactual comparison. The observed timed-search association does not authorize a universal claim about search agents or a unique causal attribution to wall-clock timing.

Fail-closed admission preserves useful negative results. Suppression does not delete mismatches, schedules, or diagnostics; it prevents an unsupported contrast from being promoted because it is favorable. Conversely, a passing preflight does not certify every future execution. The protocol records the scope in both directions.

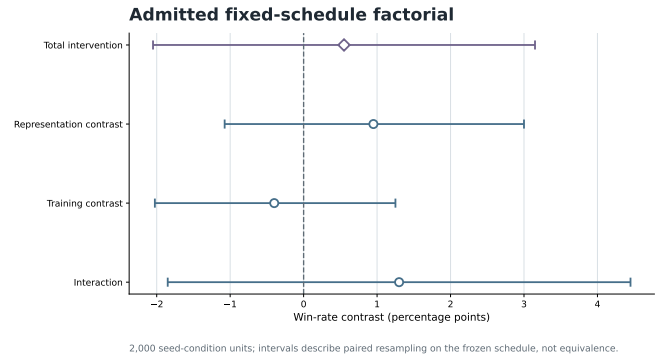


FIG. 5. Prespecified factorial contrasts and empirical 95% paired-resampling intervals for 2,000 fixed-schedule units. Intervals describe stability under the frozen resampling procedure; they are not equivalence tests or intervals over new opponents or training runs. Source data are in `source_data/figure_5_factorial.csv`.

X. LIMITATIONS

The conformance suite contains designed counterexamples and cannot estimate how often these failure modes occur. The restricted case study uses one engine build, fixed engineering opponent populations, one policy-training realization, and a declared public-observation trace projection. Opponents and execution profiles are not probability samples, so the resampling intervals do not generalize to a competition field, hardware population, or new training run.

The trace projection stores hashes of public observations rather than raw observations and omits opaque search input, private simulator state, and semantic random-event identifiers. Hash and byte-count agreement is exact for the recorded projection but cannot validate omitted state. Factorial rows contain no trace digest; their repeated-control gate compares only the frozen available record.

The engine, third-party opponents, game assets and metadata, private replay observations, policy packages, and restricted raw traces cannot be redistributed under the authority presently established. Independent readers can regenerate the synthetic suite and reaggregate the included processed diagnostics, but cannot replay the restricted trajectories end to end. Legal ownership, a redistribution license, and an archival DOI remain unresolved and therefore block any claim that the companion package is publicly available.

Finally, author identities, affiliations, ORCIDs, CRediT roles, funding, conflicts, acknowledgments, complete historical AI-use confirmation, and permissions to identify organizations or individuals require human approval. These are submission blockers, not fields to infer from repository history.

XI. CONCLUSION

A seed schedule, a repeatable execution, and an event-aligned stochastic coupling are different scientific objects. The protocol connects each object to observable evidence and to a frozen reporting consequence. Its synthetic suite demonstrates the logical failure modes and an event-keyed repair; its restricted case study shows both a failed historical gate and a passing prospective scope, followed by a bounded null-compatible factorial. The practical recommendation is to identify exact artifacts and boundary seeds, audit row parity, repeat identical arms at trace-projection level across the contexts that matter, inspect stochastic sources, and suppress any claim whose required evidence is absent.

ARTIFICIAL-INTELLIGENCE ASSISTANCE

In addition to the research-conduct uses described in Methods, OpenAI Codex assisted with bibliographic verification, manuscript drafting and editing, release-package assembly, reproducibility checklists, and adversarial review. The exact deployed GPT-5-family snapshot was not exposed. Human direction and verification procedures are recorded in `supplement/AI_USE_LOG.csv`; AI suggestions were accepted only after comparison with primary sources, code, raw or processed evidence, tests, and rendered output. No AI system is an author. The corresponding author must confirm the completeness of this disclosure and compliance with applicable tool terms, confidentiality, privacy, and intellectual-property obligations.

AUTHOR CONTRIBUTIONS

Author identities and CRediT roles for conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing, supervision, project administration, and funding acquisition require human confirmation. No role is inferred from Git history, and OpenAI Codex is not an author.

COMPETING INTERESTS

No reliable competing-interest statement is established in the repository. Every author must disclose relevant financial and nonfinancial interests, competition affiliations, employment, sponsorship, platform access, and relationships with engine or game rights holders before submission.

DATA AVAILABILITY STATEMENT

The companion review package included with this repository contains the synthetic protocol implementation and fixtures, expected outputs, processed historical and prospective diagnostics, processed factorial units, analysis and artifact code, protocols, generated figures and source data, manifests, tests, and reproduction commands. It is computationally self-contained for the synthetic and processed analyses and does not require the game engine. Public archival availability is not established: the author has not confirmed ownership, an open-source/software-data license, archive creators, maintainer contact, or a DOI. The present no-license placeholder therefore does not grant redistribution rights, and this manuscript does not promise a later release or describe the package as public.

The tournament engine and source, engine binaries, third-party opponent packages, game assets and meta-data, private replay observations, policy packages, and restricted raw traces are unavailable because of organizer terms, third-party rights, privacy constraints, and unresolved release authority. They are not offered on request because the repository does not establish legal authority or a practical controlled-access mechanism. Their identities are represented only by approved digests and bounded processed summaries. Exact restricted gameplay regeneration consequently requires separately authorized access that this article cannot provide.

Appendix A: Trace record and study-specific fields

Preflight and stress executions record schedule identity, artifact digests, the value passed at the engine boundary, outcome, errors, decision count, and a canonical trace-projection digest with byte count. The projection orders each decision’s public-observation hash, acting side, and chosen action, followed by the terminal record. Full restricted traces used for local divergence inspection are not redistributed. Factorial records intentionally use `trace_mode=none`; they contain schedule, outcome, error, and decision fields only. This distinction prevents the factorial control gate from being described as trace parity.

Appendix B: Resampling and multiplicity details

The timed-search interval resamples seed-condition clusters, retaining all four execution profiles together. The factorial intervals resample paired unit identifiers independently within each of ten opponent-by-order strata and recompute the four cell means and contrasts. Every interval uses 100,000 draws and a frozen analysis seed. The exact two-sided McNemar comparison for the total factorial contrast has 358 intervention-only wins, 347 control-only wins, and $p = 0.706483$; it is secondary and

cannot override admission. No equivalence margin was specified, so an interval containing zero cannot support equivalence.

Appendix C: Artifact and rights boundary

The synthetic conformance logic, processed summaries, source-data files, protocols, and generators are separated

from restricted engine and game materials. The release manifest rejects engine binaries, model weights, archives, credentials, absolute local paths, and non-allow-listed files. Dataset and artifact records state composition, intended verification use, and access restrictions rather than treating processed aggregates as replacements for restricted raw material [15].

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